

Paper Info

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- Title: Trust and Recall of Information across Varying Degrees of Title-Visualization Misalignment
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Why did they do this research?

The Motivation: The public desires unbiased news coverage, but current news media delivery is unsatisfactory due to news bias caused by slanted headlines. With the shift from offline to online news consumption, **clickbait headlines** are becoming more common, leading to the **faster spread of misinformation**. Although people are cautious of clickbait titles, they may be less aware of bias in data visualizations and can be influenced by the biased title of the visualization.

What did they want to learn?

Goals: The study aims to examine the issues of trust and bias with different **degrees of misalignment** between **visualization** and its **title**. Specifically, the study aims to investigate how misaligned titles affect how people recall and trust information, and how the discrepancy between the message of the visualization and the message of the title affects the perceived credibility of the data, visualization, and title. The study uses three variables to measure trust: **perceived credibility, perceived bias, and appropriateness of the title**. The research questions are formulated to explore the effects of miscued and contradictory slants in visualization titles on people's perceptions of the information.

How did they do it?

Methods: The study aimed to investigate whether people would notice the discrepancy between the title and the message in a visualization using selective slant, miscued slant, and contradictory slant (Figure 1). The study involved the recruitment of 100 participants per condition through Qualtrics, with the mean age of participants being 46 years in the miscued condition and 48 years in the contradictory condition. The study included two degrees of misalignment, miscued and contradictory, where each was created by modifying the visualization to contain misaligned messages. The study was designed with six stages, including pre-experiment attitude and demographic surveys, a preview of the visualizations and factual questions, a recall question on the main message, consistency and impartiality questions, the title recall stage, and the credibility stage. The study's results were analyzed to determine if the misalignment between the title and the visualization affected the credibility of the visualization, title, both, or none.

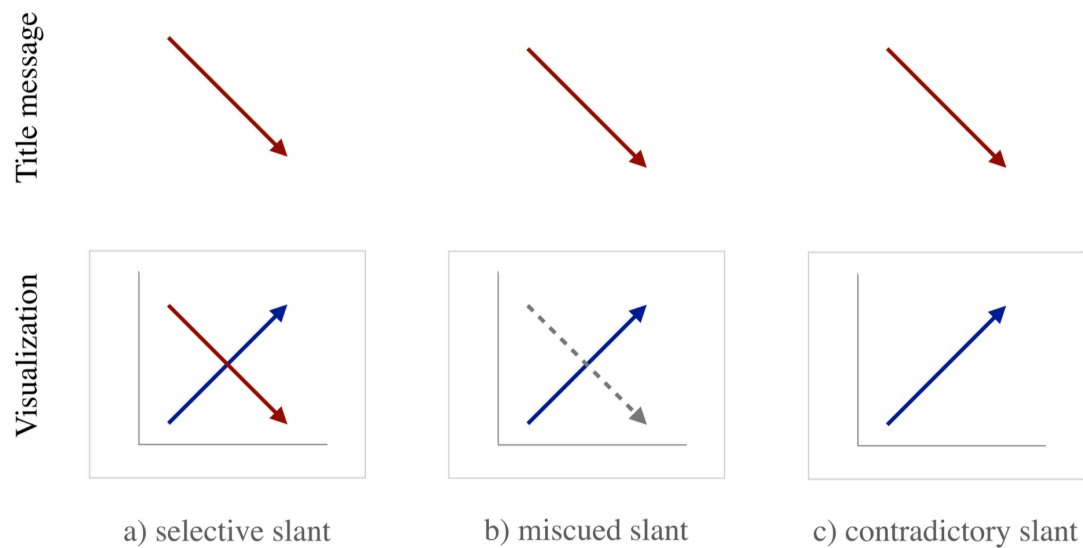


Figure 1 There are three different degrees to which the titles of visualizations can be misaligned with their intended message. Specifically, when the title highlights a decreasing trend, the visualization may a) show both trends equally, b) visually emphasize the increasing trend, or c) exclusively depict the increasing trend.

What was the conclusion?

Results: In this study, participants were shown a visualization and a title that either supported or contradicted each other in terms of the information they conveyed. The researchers found that participants tended to recall the **general topic** more often than specific messages in the title or visualization. In addition, the degree of misalignment between the title and visualization influenced participants' recall of the information, with those in the miscued condition more likely to rely on the general topic and those in the contradictory condition more likely to rely on the visualization. Surprisingly, despite the blatant slant in the title, most participants rated the information as **neutral**, with most reporting that it presented facts and statistics or that they did not have enough prior knowledge on the topic to judge its bias. However, more participants in the contradictory condition reported the military visualization as biased compared to those in the miscued condition.

My Reflection:

The study found that the misalignment between the visualization and title decreased the perceived credibility and impartiality of the information, leading to confirmation bias in interpreting and receiving information from visualizations. The results confirmed that readers heavily rely on the title while obtaining information from data, and they trust data visualization. The study also reexamines the role of text in narrative visualizations and discusses the potential of leveraging textual components to detect and combat visual-based misinformation with text-based slants. The passage suggests that textual annotations are not mere sidekicks that assist data visualizations that convey information, but they can be the storyteller with the visualization there to back up the credibility of their message.

Contributions: The article explores how visualization titles influence trust and confirmation bias when people obtain information from visualizations. The study examines an extra factor - the **misalignment between the visualization and the title** - and finds that people are more likely to

dismiss the information as less credible when the **title is attitude-inconsistent** than when the **visualization is attitude-inconsistent**. The study confirms that readers heavily rely on the title while obtaining information from data while they trust data visualization. Overall, the study provides insights into the cognitive processes involved in information recall and underscores the importance of considering multiple factors, including the alignment between titles and visualizations, and the message conveyed by titles, when designing effective communication strategies.

Limitations: The study shows that the titles have a more substantial manipulation effect than previously thought, and textual annotations can be the storyteller with the visualization there to back up the credibility of its message. The number of recalled messages aligned with the message of the titles was still greater than the number of messages aligned with the message in the visualization. The study's results confirm the hypothesis that readers heavily rely on the title while obtaining information from data while they trust data visualization.

Future follow-on work:

- Based on the findings of this study, I would like to see if there are more exploring the mechanisms behind the observed effects: The study could be complemented by additional qualitative data, such as interviews or surveys, to gain a deeper understanding of why people are influenced by visualization titles and how they interpret the information presented.
- Another potential direction is to develop guidelines for visualization design based on the study's findings, to minimize the risk of misinterpretation and improve trust and recall of visualized information. Replicating the study with visualizations from different domains, such as healthcare or finance, can help investigate whether the effects of visualization titles vary across different contexts.
- Finally, extending the study to compare the effects of visualization titles with other forms of misinformation, such as misleading captions or manipulated images, can provide insight into whether visualization titles have a unique influence on trust and recall.

Discussion Question:

1. In what ways might confirmation bias impact how people interpret visualizations, and what steps can be taken to address this issue?
2. How might the use of neutral or misleading visualization titles impact the perceived credibility of the information presented, and what strategies can be employed to avoid these issues?
3. What role do you think education and media literacy play in helping people better understand and interpret visualizations, and what can be done to improve these skills?
4. In what ways might technology, such as automated fact-checking or algorithmic detection of biased visualizations, help combat visual-based misinformation?